Deloitte. | aws

Stefan Ivic Deloitte CE Partner, AI & Data Practice Leader
Gianni Romolo AWS – Head of CEE Expansion

Deloitte-AWS Transformative Power of Cloud in Banking

Belgrade – 9th April 2025

Stefan Ivic

Deloitte CE Partner, AI & Data Practice Leader

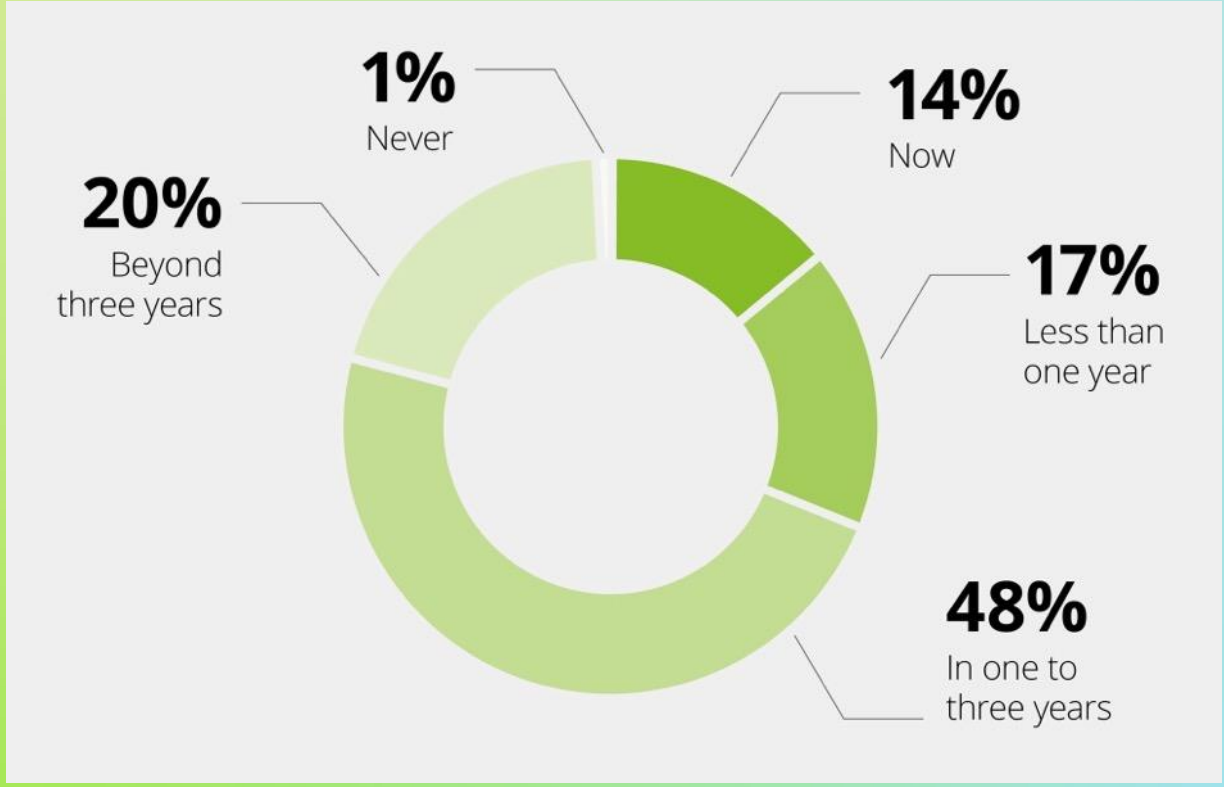
 Copyright © 2024 For information contact Deloitte Global.

Interest in Gen AI is at an all time high.

Insights from Deloitte's latest State of Generative AI in the Enterprise report:

79% of leaders expect Gen AI to transform their business within the next few years.

When do you expect Gen AI to start to transform your organization?

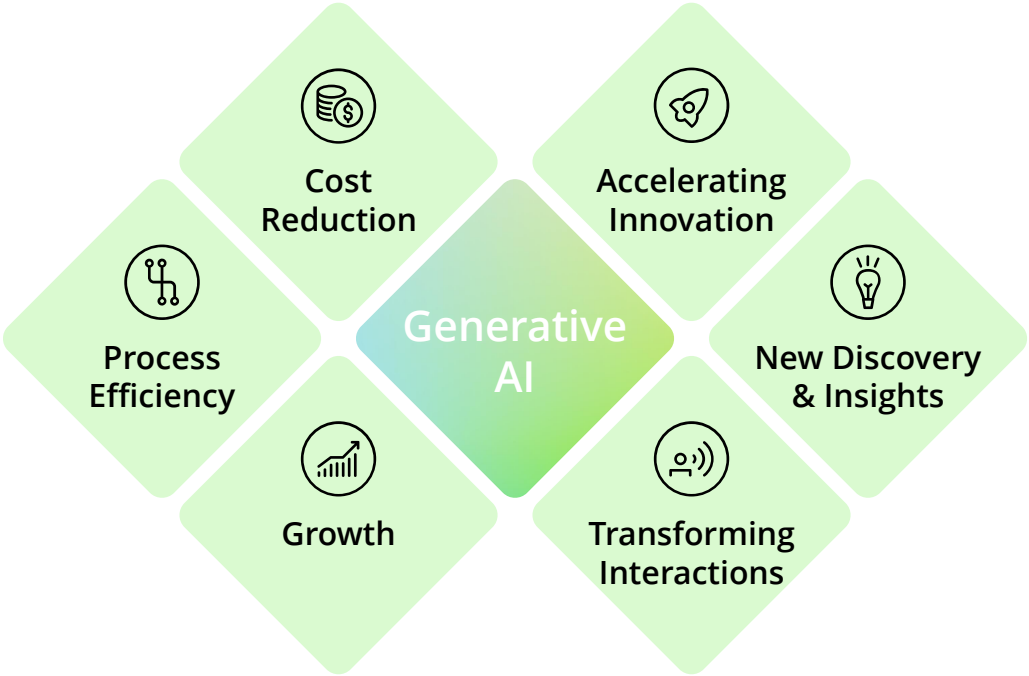


Q: When is Generative AI likely to substantially transform your organization and your industry, if at all? (Oct./Dec. 2023) N (Total) = 2,835

The unprecedented promise of Gen AI

Banks need to tackle this disruptive technology as enterprise imperative

What technology is promising to enterprises



How market dynamics are evolving

Banks perceive Gen AI as a trigger for structural transformation of their operations with broad impact ranging from cost and margin levers to innovation

Customers are already engaging with Gen AI in all aspects of life and expect similar human-centered personal finance solutions and interactions

Regulatory bodies worldwide are rolling out policies with implications for enterprises, with particular focus on compliance, data protection and confidentiality

 Copyright © 2024 For information contact Deloitte Global.

Generative AI has seized public attention

Businesses need to dive deeper into the topic to reap the benefits.

Generative AI:

- Technology that leverages machine learning to generate:
  - Text, images, audio, code, video, 3D models, and more
- Drives exponential growth in **enterprise AI adoption**
- Creating a future where:
  - **Knowledge is democratized** – accessible and understandable to all
  - **Human–machine collaboration** becomes the norm
  - **Creativity and innovation** reach unprecedented levels
  - **Boundless ideas** are generated and simulated by AI

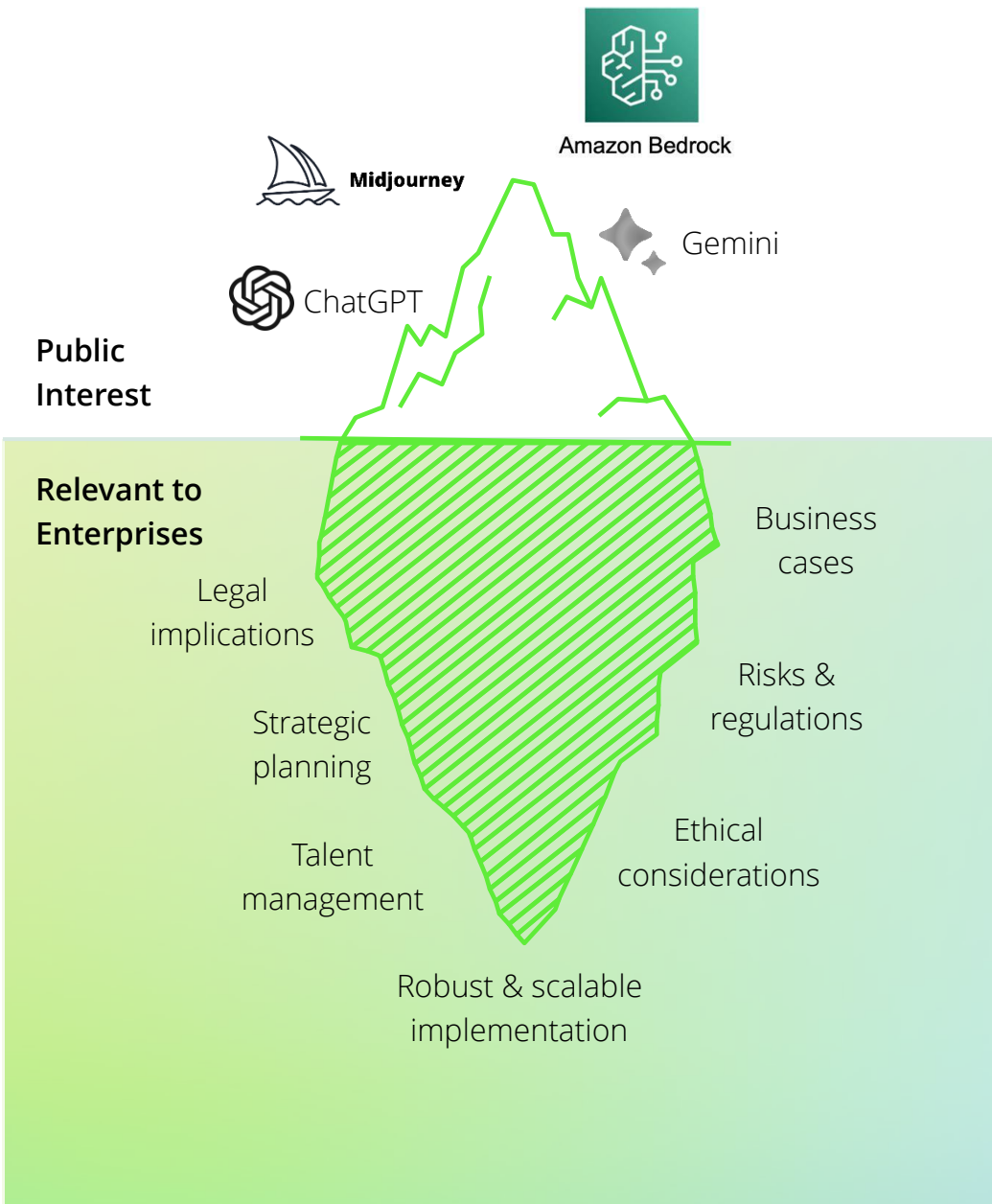
80%
of jobs will be impacted by Generative AI

+7%
increase in global GDP

+1.5pp
productivity growth

$1.3trillion
is the estimated market size for 2032 from $40B in 2022

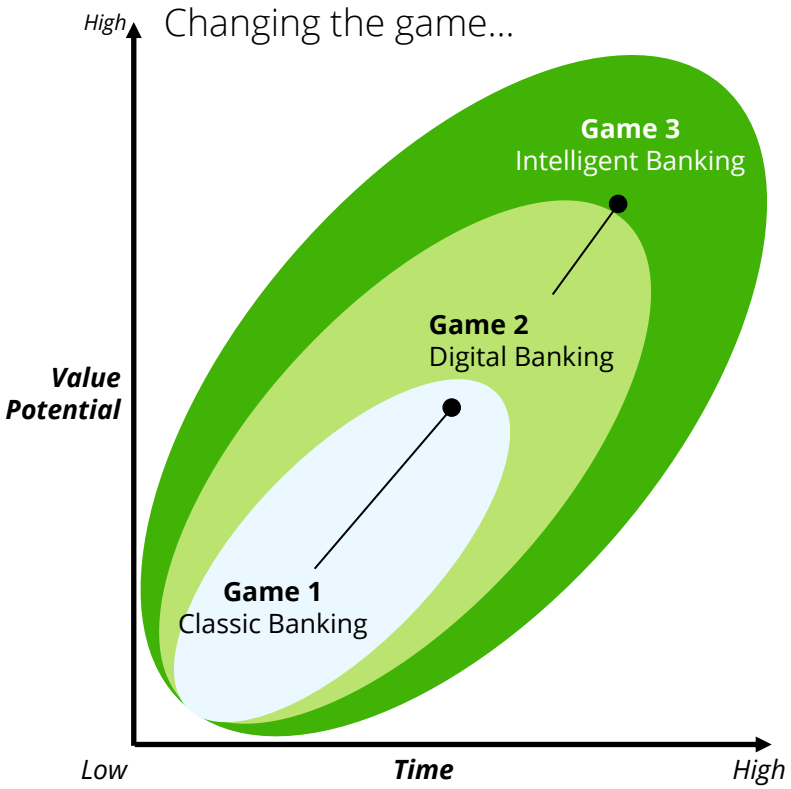
Sources: OpenAI, Gartner, Bloomberg



AI is changing the game in the Banking Industry

AI will determine the banking sector's winners and losers in the coming five years with step change towards AI first intelligent banks

How banks can capture the full value potential of AI



1) Source: Changing the game study

Game 3

- New business – strategy, segments, products, services, experiences
- Expanding the Art of the Possible (GenAI / Agentic AI)

Game 2

- New processes – same business
- Transform customer experience, personalise products/services (e.g. Digital, Data, AI)

Game 1

- Same processes – lower costs
- Steep change in efficiency and productivity (e.g. Cloud, Automation)

Key Implications

AI as a Competitive Differentiator: In a challenging economic environment, AI offers a controllable, strategic advantage for banks to remain competitive

Generative AI Expanding Use Cases: AI is opening new opportunities in banking, enhancing workforce tools and reshaping operations beyond traditional algorithmic trading

AI as a Determinant of Strategy: Banks are moving from using AI to enhance operations to redefining business models, achieving efficiency, scalability, and cost reduction

Use Cases

Reducing fraud cases: NatWest cut fraud by 6%, with a 90% drop in account opening fraud since 2019, lowering operational costs

Loans underwriting: UK banks have been fully automating the loans underwriting process up to US$100,000

Productivity increase: Projected 27% employee productivity increase across investment banks by 2026

 Copyright © 2024 For information contact Deloitte Global.

GenAI Powered Transformation Targets 3 Key Value Plays That Can Create Competitive Advantages

EFFICIENCY PLAY

Improve operational performance through improved productivity by doing more with less

- Automate processes
- Optimize costs
- Drive consistency
- Reduce FTEs
- Remove waste
- Improve speed

Profitability and Margin Improvement
Cost : Income Ratio

Stakeholder Value
EPS, EBITDA and Revenue

Product & Portfolio Performance
Profitability, P&Ls, Margins

EXPERIENCE PLAY

Provide fit-for-purpose and customized experiences to customers and employees

- Personalize content
- Differentiate services
- Enhance Quality and Outcomes
- Amplify creativity
- Knowledge Shift

Engagement Centricity
Customer Journey Engagement

Outcomes & Indicators, Adherence
Banking Reported Outcomes, Adherence

New Digital Products & Services
New integrated offerings

CAPABILITY PLAY

Develop and enhance enterprise digital and data capabilities that are enabled by GenAI

- Generate new insights
- Extend Asset Utility
- Expand Workflow skills
- Improve decisioning
- Improve adaptability

Future Proof Technologies
Pivotal, Fit-for-Purpose Architectures

Business Model Agility
Time to Market

Workforce & Workplace of the Future
Virtual, Automated, Augmented Workforce

 Copyright © 2024 For information contact Deloitte Global.

The banking industry is buzzing with activity

EXPERIENCE PLAY

- Transforming Wealth Management with AI Chatbots — Morgan Stanley
- Product suggestions for investment — Deutsche Bank
- GenAI for Software Development and Information Analysis — CITADEL
- Automating Document Processing and Scaling Virtual Assistants — WELLS FARGO
- Cosmos.AI: Enables tasks like fraud detection and personalized customer service — PayPal

EFFICIENCY PLAY

- Named entity extractors for sentiment analysis — Bloomberg
- Enhancing Employee Efficiency and Customer Service — OCBC
- IndexGPT: AI service for investment advice — JPMORGAN CHASE & CO
- Advanced Investment Strategies and Risk Analysis with Gen AI — BlackRock
- Enhancing Fraud Detection using GenAI — Capital One

CAPABILTY PLAY

- Access creditworthiness of borrowers — Upstart
- AML transaction monitoring & sanctions screening — HSBC
- Enhancing User Experience and Security with Gen AI — Revolut
- Streamlining Banking Operations through GenAI — Hokuhoku Financial Group, Inc.
- AI for cybersecurity — Goldman Sachs
- Digital Assistant for retail banking customers — NatWest
- Automated analytics and decision-making — citibank
- NOMI: Help clients manage everyday finances — RBC Royal Bank of Canada

AI solutions will rapidly evolve towards autonomous agents (Agentic AI)

AI's evolution will empower automated problem detection and resolution, enhancing efficiency and accuracy

Sector Impact of GenAI on Banking & Capital Markets

Stage 1

AI ASSIST

10-20% potential

Personal agent focused on productivity: Executive and specialist roles e.g. functional leaders, top levels of management

Use case:

Assistants/Chatbots: Automates a specific task like transcribing notes from a meeting or asking for generation information about a topic

Stage 2

AI AUGMENTATION

20-50% potential

Specialist agent focused on improving quality: Those with domain knowledge e.g. investment manager, underwriters, relationship/account managers

Use case:

Advisor: Provides context, creates insights and makes suggestions like the automated A/B testing for marketing campaigns with recommendations

Stage 3

AI AUTOMATION

50-80% potential

Transforming process focused on cost reduction: Customer facing and support roles e.g. contact center agents, central services

Use case:

Autonomous Agents: Understands, plans and takes action autonomously like security threat modelling, identification and remediation

 Copyright © 2024 For information contact Deloitte Global.

Considerations for identifying application areas of AI in Banking

Success in AI can be built on the right technical foundation paired with strong execution

Who will lead the cohort

"Banks that led prior tech innovations (cloud, digital, data) will outperform in AI-driven value creation.

Their investments in culture, governance, and agile methods have laid the groundwork.

While past investments limited short-term gains, AI now acts as the catalyst to unlock their full potential…"

& what they need to be successful

Banks need to be pioneers in key technology …

… but also develop an AI driven organization

AI Strategy & AI Ambition

+ Use Case Definition & Roadmap

Technology

- **Cloud & Infrastructure**
- **Data Management**
- **AI Ecosystem & Assets**

Deloitte. | aws

Scalable computing power and accessible quality data provisioning reduce IT complexity and lower costs while enabling faster AI deployment

Organization

- **Legal, Risk & Compliance**
- **Operating Model & Governance**
- **People Transformation**

Deloitte.

- Robust frameworks to mitigate regulatory risks, ensure responsible AI usage, and maintain compliance.
- A culture of innovation, upskilling talent, and fostering cross-functional collaboration with streamlined workflows and governance models

MLOps Delivery Lifecycle (Execution) and Operations

 Copyright © 2024 For information contact Deloitte Global.

Timeline for driving value and scale

With our clients, we typically see an ~18-month journey to driving value and scale.

1 Strategize and mobilize (3 months)

- Develop an **AI strategic blueprint**
- Mobilize **stakeholders**
- Identify **high-value** transformation opportunities

Result: AI North Stars and governance framed, key use cases identified

Deloitte. | aws

2 Foundational bets (6 months)

- Deliver **quick wins** from efficiency plays
- Develop strategic **partner ecosystem**
- Develop **enterprise controls for** responsible adoption

Result: MVPs developed; key technology and governance decisions made

Deloitte. | aws

3 Iteratively scale (6 months)

- **Scale MVPs** and **embed AI** into business workflows
- **Measure outcomes** from adoption
- Build **AI fluency** and **upskill** workforce

Result: AI savvy workforce, initial value realization

Deloitte.

4 Institutionalize AI (ongoing)

- Deliver new use cases to **complete end to end transformation** of value streams
- Develop **enterprise AI assets and capabilities** to unlock efficiencies

Result: Scaled business transformation

Deloitte.

 Copyright © 2024 For information contact Deloitte Global.

Gianni Romolo

AWS – Head of CEE Expansion

 Copyright © 2024 For information contact Deloitte Global.

Why customers choose AWS

EXPERIENCE

18 years

helping millions

GLOBAL REACH

36 Regions

spanning 114
Availability Zones and

SECURITY

300+

security features

INNOVATION

250+

service offerings

AWS INFRASTRUCTURE

3.6x+

energy efficient than the
median of surveyed U

TCO

151

price reductions
since 2006

ECOSYSTEM

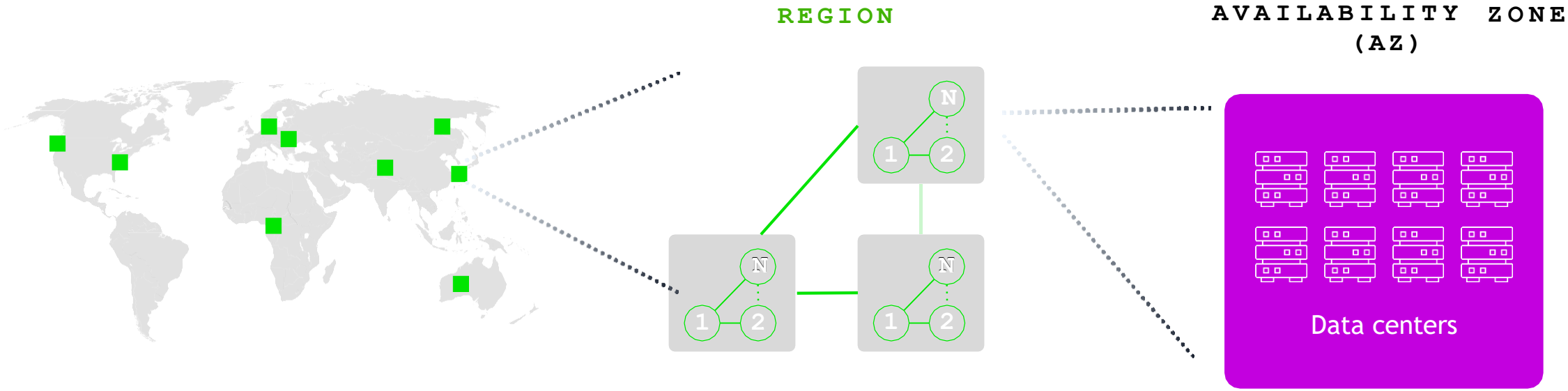
12,000

software listings from
2,000 ISVs

Source: AWS Approved Public Stats

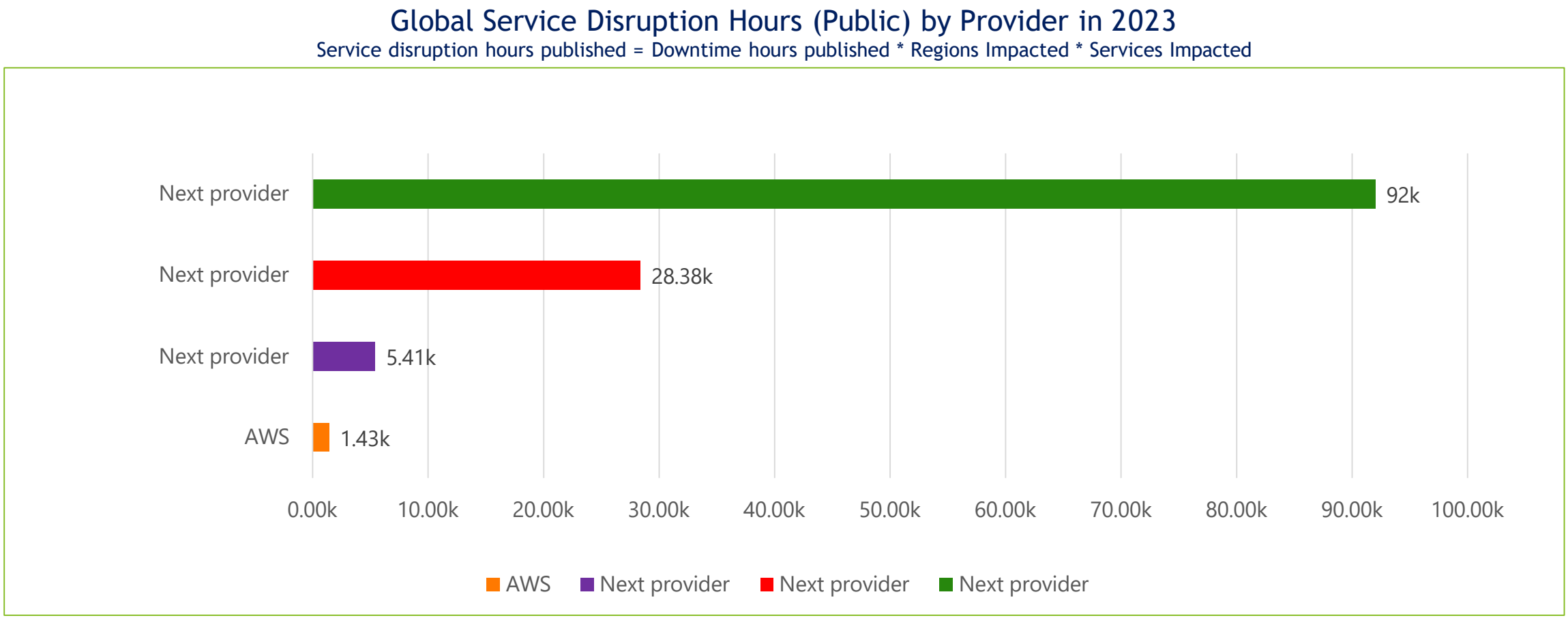
AWS Region design

AWS Regions are comprised of multiple AZs for high availability, high scalability, and high fault tolerance. Applications and data are replicated in real time and consistent in the different AZs.



AWS Cloud Infrastructure Designed to Scale with Highest Availability

AWS Investments on Underlying Higher Resilient Infrastructure translates into lower costs to customers



* Sources: Public Status Pages for Cloud Providers

Deloitte. | aws

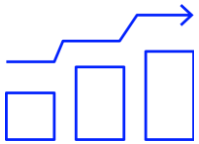
Commercial Banks

Commercial Banks are using AWS to drive their transformation and innovation agendas

Modernizing the customer experience with digital channels and better Relationship Manager enablement

Embedding AI/ML in operations to enhance productivity, improve document processing, and reduce fraud

Building API ecosystems to drive efficiencies and ease integration



Running on-demand risk calculations with scalability

Standard Chartered cut its risk grid costs with AWS

> **“Using Amazon EC2 Spot Instances, we can experiment and test more…We can check performance of different hardware types with a single click, as opposed to the traditional months-long testing processes, making it even easier to provide our customers with the best-performing services.”**

Richard Davis
Global Head of Technology Services, Standard Chartered

OVERVIEW

Standard Chartered needed to run complex computer model in order to establish comprehensive risk pictures and comply with the Fundamental Review of the Trading Book (FRTB) and other regulatory requirements.

SOLUTION

The bank chose AWS because of its leading position in the market and the breadth and maturity of services available.

IMPACT

By moving to AWS, Standard Chartered tripled its compute capacity and reduced its compute costs by 60% by moving to Amazon EC2 Spot Instances. The bank now uses 70x more compute resources on AWS than it had access to on-premises and has taken a cloud-first approach to all software development.

AWS is the fabric that connects an expansive partner community

Consulting Partner

Deloitte.

Core Banking Platforms

10x

Infosys Finacle

MAMBU

tcs BaNCS

avaloq

Finxact CORE AS A SERVICE

[redacted]

temenos

Customer Experience

Customer 360 / Analytics

qualtrics XM

GLASSBOX

Amplitude

Digital Engagement

intellect Design for Digital

perx

affirm

ncino

MONEYTHOR

CRM

CRMNEXT

salesforce

HubSpot

Chatbots

clinc

personetics

Fraud / KYC / AML

ID Verification

jumio

Socure

veriff

onfido

Lifecycle Management

NICE ACTIMIZE

fenergo

FICO

Transaction Monitoring

COMPLY ADVANTAGE

FIS

feedzai

Fraud.net

Data Enrichment

dun & bradstreet

REFINITIV

Lending

behalf

experian

Moody's ANALYTICS

Finxact CORE AS A SERVICE

OakNorth

ncino

Accounting / FP&A

Aptitude

SAP

workday

xero

Deloitte. | aws

Retail & Small Business (SME) Banking

We support our banking customers in a variety of ways

Industry affiliations

Relationships with leading industry associations, including Sibos and FS-ISAC, as well as a program of AWS-led industry events

Compliance best practices

Enablers to help customers address regulatory frameworks and maintain a compliance-ready environment, including FFIEC Workbook & Audit Guides

Professional Services

Specialists with industry and technology experience that customers engage to define and implement a migration or compliance program

Training & certification

Tailored technical, security, compliance, and business training to accelerate cloud adoption

AWS FS Partner Competency

Global program comprising Financial Services technology and consulting companies that can help customers shift critical workloads to the cloud and comply with industry regulations

In-house industry expertise

Resources include former regulators, compliance officers, information security experts, governance and audit professionals, traders, and technology specialists with industry experience

Deloitte. | aws

Digital Native Banking

Cloud-based digital banking is transforming the industry

Speed to market

By leveraging the cloud, digital banks can bring new products to market faster than their traditional counterparts, changing the economics of banking and enabling international growth and diversification of profit.

Enhanced customer experiences

Digital banks use cloud-based AI/ML tools to gather deep customer insights and provide personalized experiences.

Evolving bank regulations are encouraging competition

Regulators are encouraging innovation and competition through open banking and open finance, and by granting digital-only banking licenses.

There are three main paths to launching a digital bank

Challenger Banks/FinTechs

Independent start-up banks with a banking license

Digital Bank Offshoots/"Speedboats"

Subsidiary or extension of an existing incumbent bank

Build

Use AWS native services to build core banking systems that meet your needs

Buy & Build

Build with Banking SaaS providers on AWS infrastructure

Buy

Use Banking-as-a-Service providers on AWS

Raiffeisen Digital Bank is getting closer to its customers

OVERVIEW

Raiffeisen Digital Bank is getting closer to its customers

SOLUTION

Raiffeisen Bank International built Raiffeisen Digital Bank (RDB), an independent neobank by leveraging on Mambu on AWS for its core systems, an AWS Data Lake as the center piece of its back-end services, and AWS Managed Containers.

IMPACT

RDB grew its lending portfolio to 50M Euros in 18 months; can make a lending decision in 15 minutes or less from app download, and disburse money for approved loans in under 30 minutes.

Featuring AWS Financial Services partner

engine

Salt Bank builds a bank in 12 months on AWS with Engine

OVERVIEW

Salt Bank wanted to provide its customers with digital-native, mobile-first experiences. They wanted to disrupt Romanian banking and put customers first by offering the agility of a fintech with the safety and security of being a licensed, regulated bank.

SOLUTION

The bank chose to use Engine by Starling as its core banking provider. Salt wanted the bank to be cloud-native. They chose to run it on AWS due to Engine's extensive experience working with AWS, and AWS's proven ability to enhance availability and agility, enable rapid development and maintain high security.

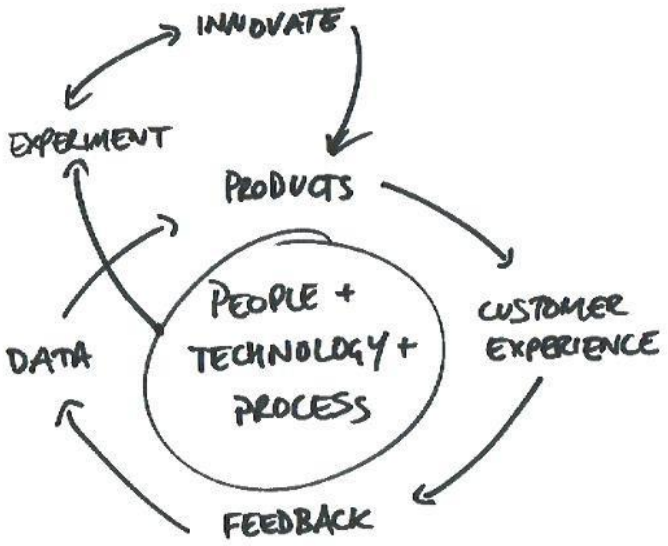
IMPACT

Salt now offers transaction accounts in local and 16 foreign currencies as well as automated digital onboarding. Customers have access to innovative, self-serve banking features such as savings, spending insights, and cost controls. The bank already has 260,000 active customers since launch in April 2024 and is currently the #1 finance app on Apple and Google Play stores in Romania.

Featuring AWS Financial Services partner

engine

Deloitte. | aws



Art of Possible *Workshop*

- **A 1-2 day workshop based wholly and solely on figuring out the key needs of the customer.**
- Creating a Data Driven strategy that provides a discrete customer experience.
- Interaction with leaders to find and solve the key issues using Deloitte and AWS technological prowess.

The workshop provides the foundation, utilizing Amazon's working backwards approach, for a 30-60-90-120 innovative project or initiative implementation sprint plan with well-defined milestones to quickly demonstrate results.

Deloitte. | aws

Visit us at out booth!

GenAI within Banking

The art of possible in Banking with latest GenAI technologies

Trends - and Business Value of Cloud in Banking

Focus of technology spending in Banking 2025-2030 and business value cloud brings at the transformation

Banking Specific Demos

- Intelligent Document Processing
- Mortgage Assistant
- Loan Assistant
- Dashboard Analytics with Amazon Q

Gianni Romolo ☁️

Head of CEE Expansion - Bridging Digital Transformation and Finan...

Stefan Ivic

AI & Data Practice Leader for Deloitte Central Europe

THANK YOU!